Análisis Matricial de Pórticos Espaciales

Modelado 3D Completo: 6 GDL por Nudo

Referencia: Kassimali — Matrix Analysis of Structures, Cap. 7

1. Propiedades del Sistema

Se han definido propiedades constantes para los elementos prismáticos:

- **Módulo de Elasticidad (E):** 200000000 kPa
- **Módulo de Cortante (G):** 80,000,000 kPa
- **Área (A):** 0.0132 m^2
- **Momentos de Inercia (I_x, I_y):** 0.0004, 3.5e-05 m^4
- **Constante de Torsión (J):** 1.5e-06 m^4

2. Formulación de la Matriz de Rigidez 3D

La matriz de rigidez local $\boldsymbol{k}$ del elemento de pórtico espacial es de 12×12. Integra los efectos de rigidez axial, torsión y flexión en dos planos perpendiculares:

$$\boldsymbol{T} = \begin{pmatrix} \boldsymbol{\Gamma} & 0 & 0 & 0 \\ 0 & \boldsymbol{\Gamma} & 0 & 0 \\ 0 & 0 & \boldsymbol{\Gamma} & 0 \\ 0 & 0 & 0 & \boldsymbol{\Gamma} \end{pmatrix} \qquad \boldsymbol{\Gamma} = \begin{pmatrix} c_x & c_y & c_z \\ l_x & l_y & l_z \\ m_x & m_y & m_z \end{pmatrix} \tag{1}$$

3. Matriz de Rigidez Global del Sistema (S)

La matriz global tiene una dimensión de 24 ×24. Debido a la magnitud de los datos en 3D, se presenta una sección de la matriz con escala reducida:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
0	440777.8	0.0	0.0	0.0	1166.7	1166.7	−388.9	0.0	0.0	0.0	1166.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	448888.9	0.0	−13333.3	0.0	13333.3	0.0	−4444.4	0.0	−13333.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	444833.3	−13333.3	−1166.7	0.0	0.0	0.0	−440000.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	−13333.3	−13333.3	106686.7	0.0	0.0	0.0	13333.3	0.0	26666.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	1166.7	0.0	−1166.7	0.0	9353.3	0.0	−1166.7	0.0	0.0	0.0	2333.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	1166.7	13333.3	0.0	0.0	0.0	58020.0	0.0	0.0	0.0	0.0	0.0	−20.0	0.0	0.0	0.0	0.0	0.0	0.0
6	−388.9	0.0	0.0	0.0	−1166.7	0.0	440777.8	0.0	0.0	0.0	−1166.7	1166.7	−440000.0	0.0	0.0	0.0	0.0	0.0
7	0.0	−4444.4	0.0	13333.3	0.0	0.0	0.0	448888.9	0.0	13333.3	0.0	13333.3	0.0	−4444.4	0.0	0.0	0.0	13333.3
8	0.0	0.0	−440000.0	0.0	0.0	0.0	0.0	0.0	444833.3	−13333.3	−1166.7	0.0	0.0	0.0	−388.9	0.0	−1166.7	0.0
9	0.0	−13333.3	0.0	26666.7	0.0	0.0	0.0	13333.3	−13333.3	106686.7	0.0	0.0	0.0	0.0	0.0	−20.0	0.0	0.0
10	1166.7	0.0	0.0	0.0	2333.3	0.0	−1166.7	0.0	−1166.7	0.0	9353.3	0.0	0.0	0.0	1166.7	0.0	2333.3	0.0
11	0.0	0.0	0.0	0.0	0.0	−20.0	1166.7	13333.3	0.0	0.0	0.0	58020.0	0.0	−13333.3	0.0	0.0	0.0	26666.7
12	0.0	0.0	0.0	0.0	0.0	0.0	−440000.0	0.0	0.0	0.0	0.0	0.0	440777.8	0.0	0.0	0.0	−1166.7	1166.7
13	0.0	0.0	0.0	0.0	0.0	0.0	0.0	−4444.4	0.0	0.0	0.0	−13333.3	0.0	448888.9	0.0	13333.3	0.0	−13333.3
14	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	−388.9	0.0	1166.7	0.0	0.0	0.0	444833.3	−13333.3	1166.7	0.0
15	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	−20.0	0.0	0.0	0.0	13333.3	−13333.3	106686.7	0.0	0.0
16	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	−1166.7	0.0	2333.3	0.0	−1166.7	0.0	1166.7	0.0	9353.3	0.0
17	0.0	0.0	0.0	0.0	0.0	0.0	0.0	13333.3	0.0	0.0	0.0	26666.7	1166.7	−13333.3	0.0	0.0	0.0	58020.0

4. Desplazamientos y Rotaciones Nodales

	Nodo	u_X	u_Y	u_Z	θ_X	θ_Y	θ_Z
0	1	0.0	0.0	0.0	0.0	0.0	0.0
1	2	0.0	0.0	0.0	0.0	0.0	0.0
2	3	0.0	0.0	0.0	0.0	0.0	0.0
3	4	0.0	0.0	0.0	0.0	0.0	0.0
4	5	−0.139	−0.000704	0.0154	0.00342	0.0257	−0.00287
5	6	0.139	−0.000659	0.0154	−0.000344	0.0257	−0.00671
6	7	0.139	−0.000704	−0.0154	−0.00342	0.0257	0.00286
7	8	−0.139	−0.000659	−0.0154	0.000344	0.0257	0.00671

5. Fuerzas Locales en los Miembros

	Elemento	Axial (N)	Torsor (T)	Mz_i (kNm)	Mz_j (kNm)
0	1	309.91	−0.51	−114.72	−23.64
1	2	290.09	−0.51	−214.21	−223.39
2	3	309.91	−0.51	114.72	23.64
3	4	290.09	−0.51	214.21	223.39
4	5	105.52	0.06	175.40	130.95
5	6	−94.36	0.06	−130.73	−175.62
6	7	24.94	0.08	−23.58	−223.33
7	8	24.94	0.08	−23.58	−223.33

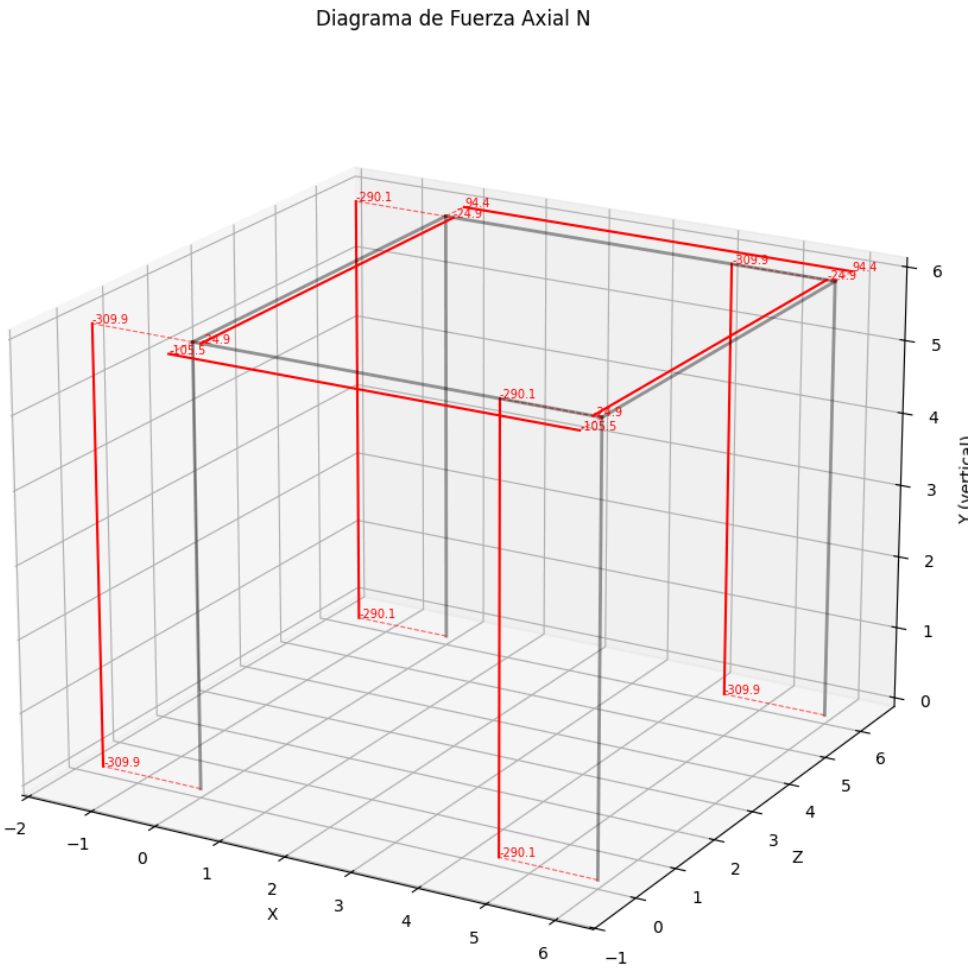

Figura 1: Diagrama de Fuerza Axial (N)

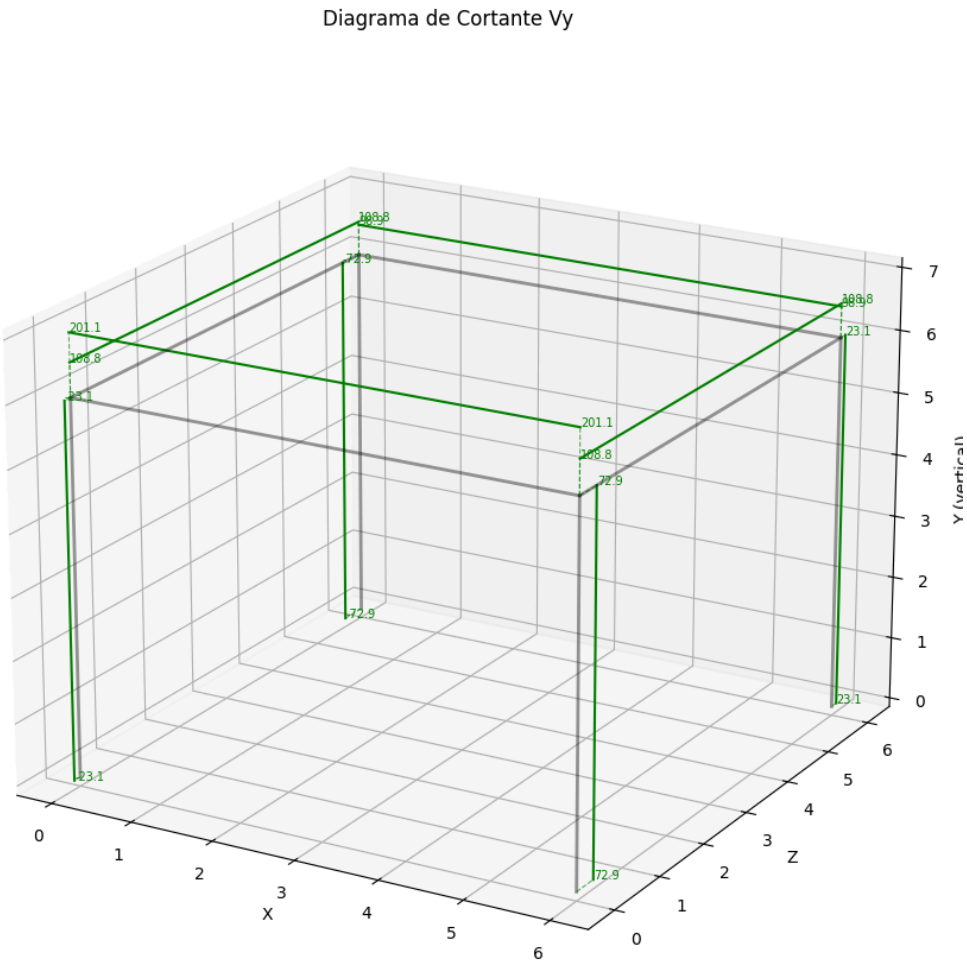

Figura 2: Diagrama de Fuerza Cortante V_y

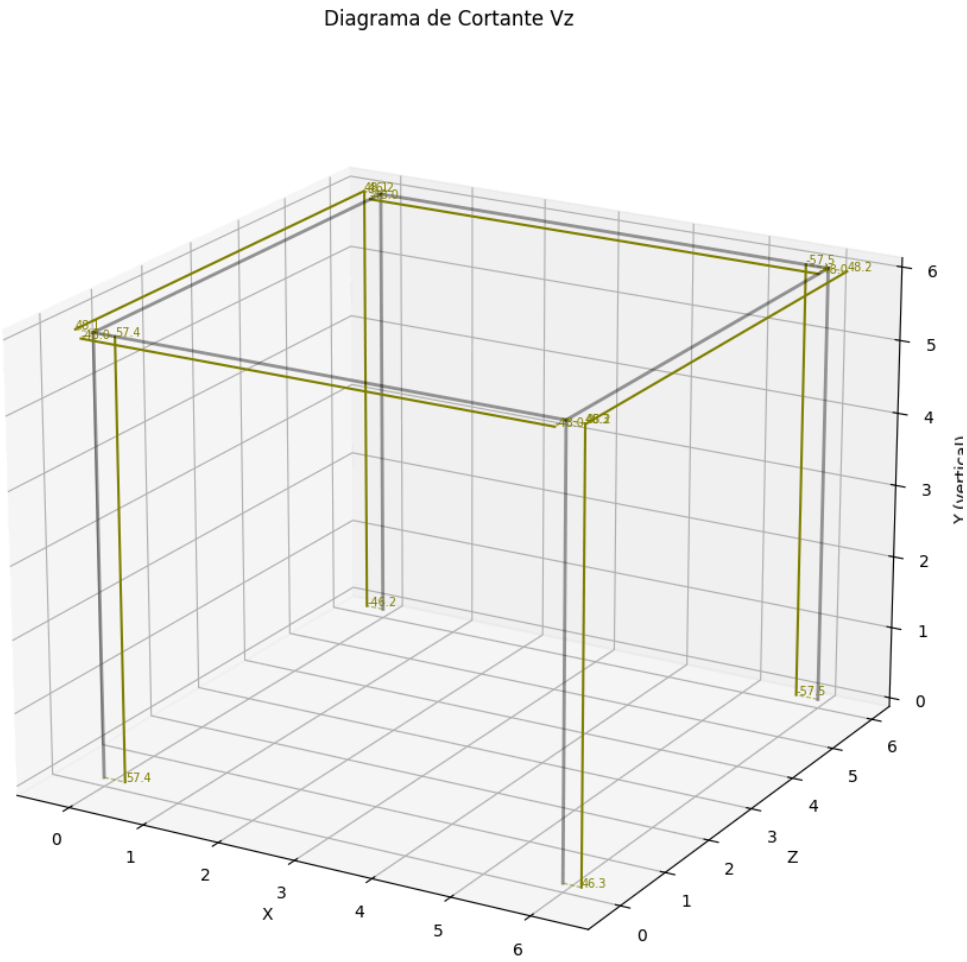

Figura 3: Diagrama de Fuerza Cortante V_z

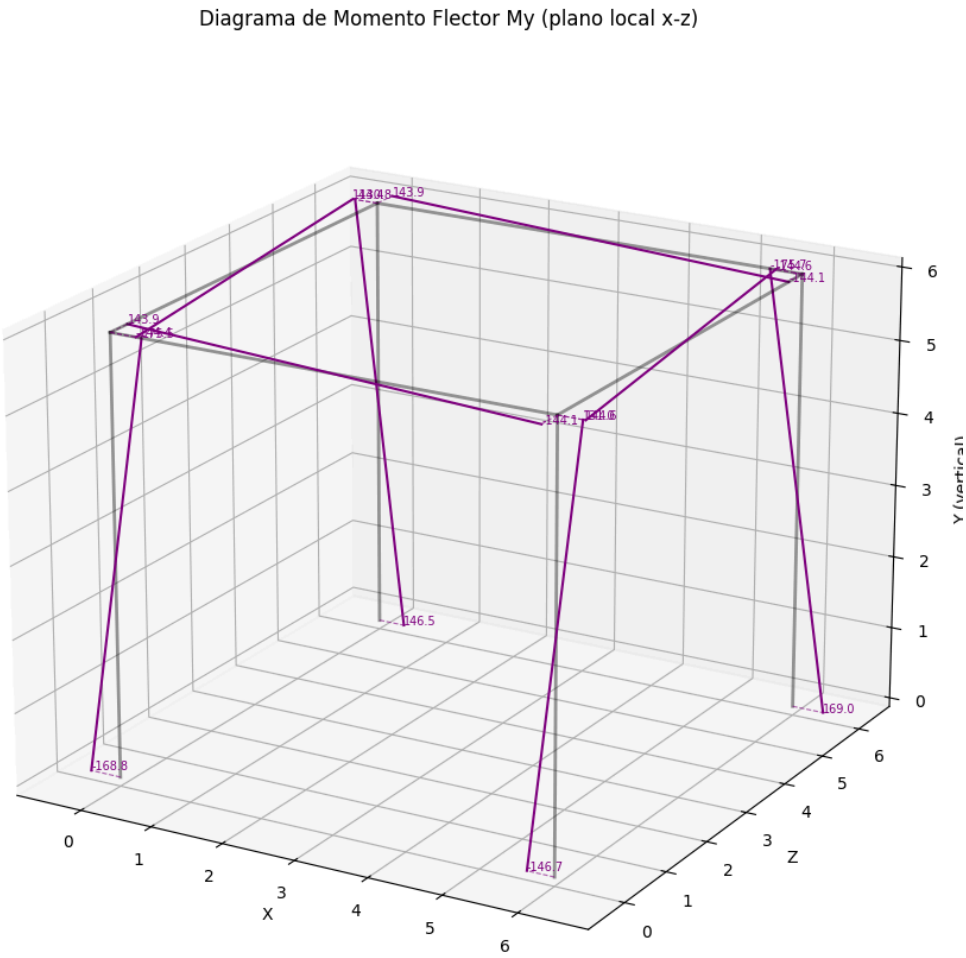

Figura 4: Diagrama de Momento Flector M_y

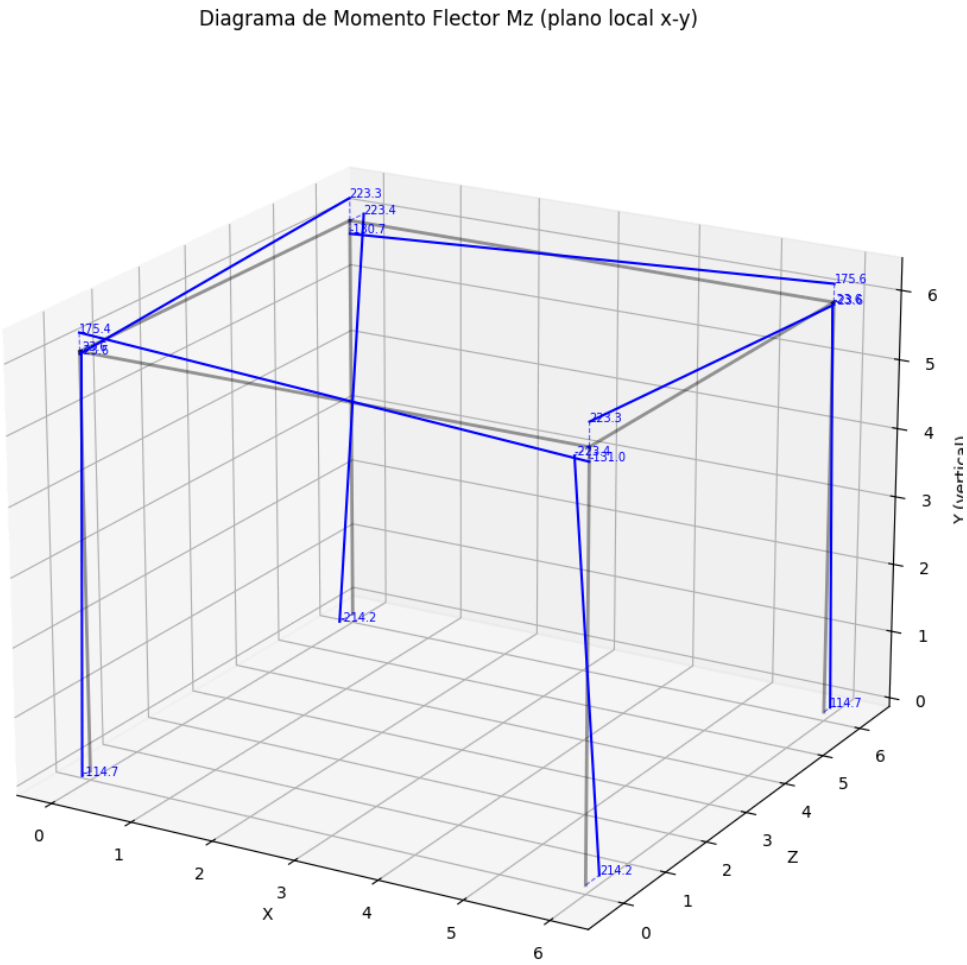

Figura 5: Diagrama de Momento Flector M_z

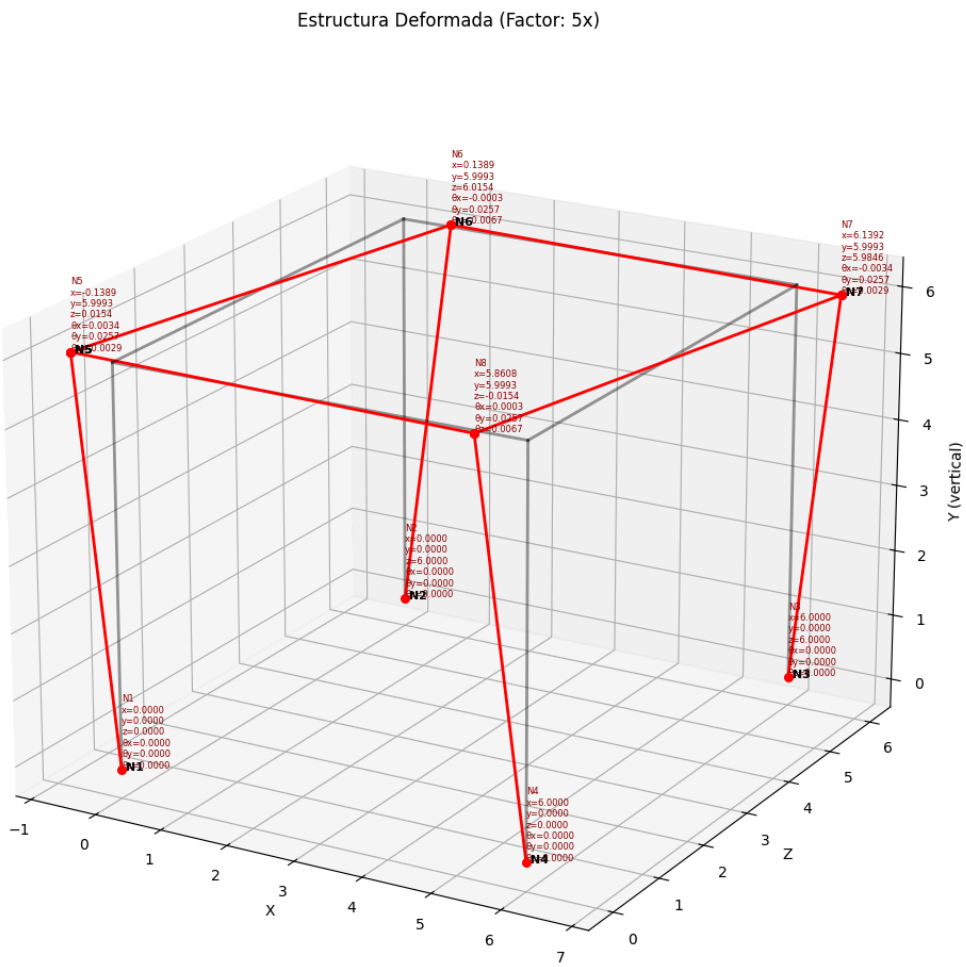

Figura 6: Vista de la Estructura Deformada

6. Acciones Finales en Nudos (Coordenadas Locales)

	ID	N (kN)	Vy (kN)	Vz (kN)	My (kNm)	Mz (kNm)
0	1	309.91	−23.06	57.38	−168.78	−114.72
1	2	290.09	−72.93	−46.21	146.45	−214.21
2	3	309.91	23.06	−57.45	169.02	114.72
3	4	290.09	72.93	46.29	−146.70	214.21
4	5	105.52	201.06	−48.00	143.93	175.40
5	6	−94.36	98.94	−48.00	143.93	−130.73
6	7	24.94	108.85	48.15	−144.44	−23.58
7	8	24.94	108.85	48.19	−144.57	−23.58